

Fig. 3: Pin details of 433MHz transmitter and receiver

oscillator formed by gate N7, C6 and R10 are fed to pin 2 of gate N5. The pulse output at pin 3 of gate N5 is remodulated with the 38kHz subcarrier to provide noise immunity in IR remote control systems.

These 38kHz tone bursts are then fed through buffer inverting gate N8 to transistor T5 via base resistor R7. Thus T5 is driven at a 38kHz rate for each of the received code pulses. The IR transmitting LEDs (IR LED1 and IR LED2) produce the regenerated IR control pulses for your DVD player. At the same time, LED2 flashes to

indicate reception of the transmitted signal. Diode D4 prevents damage due to accidental reverse-polarity connection of the plug-pack supply, while C9 provides additional smoothing.

Both the transmitter and receiver units are designed to run from an adaptor or any other convenient 12V DC source.

Assemble both the units on separate general-purpose PCBs and enclose in suitable plastic cabinets. Refer Fig. 3 for the pin details of the transmitter and receiver. After completing the assembly, solder an antenna at an appropriate place on each of the PCBs.

Before installation, check the

transmitter and receiver for proper working. To check the transmitter unit, supply power via its plug-pack. LED1 may blink very briefly when the power is applied, but then it should go off until you point your remote to its front window and press one of its buttons. LED1 of the transmitter unit should blink to confirm that the transmitter is receiving the signal from the IR remote.

To check the working of the receiver, assume that the transmitter is working well. Now power-on the receiver as well. Then with the transmitter and receiver units a metre or two apart (but not facing each other), point your remote to the front of the transmitter again—this time watching the front of the receiver. The LED2 of the receiver should blink, indicating that the receiver is receiving the codes correctly. ●

## TELEPHONE LINE IN-USE INDICATOR

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This circuit gives a visual indication of whether a telephone line is in use or not, helping two or more people sharing the line to avoid interrupting each other's calls. As the circuit connects in parallel with the telephone line, it can also detect phone tapping/misuse attempts. The circuit draws negligible current from the telephone line for its operation.

Fig. 1 shows the circuit of the telephone line indicator. When the telephone is not in use (the handset is on hook), the line voltage is 48-60V. This voltage passes through the simple polarity guard built around the bridge rectifier comprising diodes D1 through D4 and fed to the remaining circuit. Zener diode ZD1 breaks down and transistor T1 conducts. As a result, transistor T2 does not conduct and LED1 does not glow, indicating that

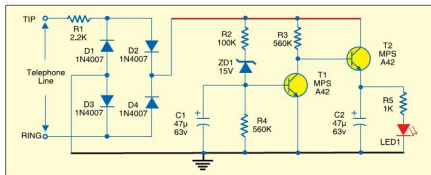


Fig. 1: Telephone line indicator circuit



the line is not in use.

When the telephone line is in use (the handset is off-hook), the line voltage drops to a level below the zener voltage of ZD1. Transistor T1 cuts off and transistor T2 conducts. As a result, LED1 lights up to indicate a busy/in use/tap/misuse condition.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Refer Fig. 2 for the pin configuration of transistor MPSA42 and LED. ●

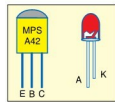


Fig. 2: Pin configurations of MPSA42 and LED